



AIMS10-01-005

Issue 3

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AIMS
Airbus Material Specification

Structural Adhesive Systems-Film
Epoxy, for composite bonding
(120°C performance)
180°C curing class

Material Specification

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1 Scope

This Material Specification defines the requirements of qualification of structural adhesives for composite bonding, supplied in the form of a film supported by a carrier, curing at 180°C.

This specification shall be used in conjunction with AIMS10-01-000 Part 1 and AIMS10-01-000 Part 3 for aerospace application.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be applied in conjunction with the referenced documents.

2 Application

Structural bonding of composites, composites to honeycomb cores and composites to other adherents including, co-bonding and co-curing processes at 180°C. The adhesives films have a temperature range of -55°C to 120°C. Grade A adhesives are suitable for composite to composite applications. Grades B and C adhesives are suitable also for composites to honeycomb core applications.

3 Normative references

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS10-01-000 Part 1	Airbus Technical Specification - Structural Adhesive Systems-Film, Part 1 General Requirements
AIMS10-01-000 Part 100	Structural Adhesive Systems - Films Acceptance Testing and Storage Life Prolongation - Technical Specification
AIMS10-01-000 Part 3	Technical Specification Structural Adhesive Systems Part 3: Qualification Program / Batch Release Testing Film for composite bonding
AITM 1.0003	Determination of the glass transition temperature
AITM 1.0019	Airbus Test Method - Determination of Tensile Lap Shear Strength of composite joints
AITM 1.0025	Airbus Test Method – Fiber Reinforced Plastics – Flatwise tensile test of composite sandwich panel
AITM 3.0001	Analysis of thermoset systems by high performance liquid chromatography
AITM 3.0002	Analysis of non-metallic material (uncured) by differential scanning calorimetry
AITM 3.0003	Analysis of organic compounds by infrared spectroscopy
AITM 3.0008	Determination of the extent of cure by Differential scanning calorimetry
EN 2243-1	Aerospace series - Structural adhesive -Test methods - Determination of single lap shear strength
EN 2243-2	Aerospace series - Structural adhesives - Test methods - Peel metal-metal
EN 2243-6	Aerospace series - Structural adhesives - Test methods - Determination of shear stress and shear strain
EN 2243-7	Aerospace series - Structural adhesives - Test methods - Determination of the flow characteristics of a film adhesive
EN 2558	Aerospace series - Carbon fibre preimpregnates - Determination of the volatile content
EN2243-3	Non-metallic materials - Structural adhesives - Test method - Part 3 :Peeling test metal-honeycomb core
EN2243-4	Non-metallic materials - Structural adhesives - Test method - Part 4 : Metal-honeycomb core flatwise tensile test
EN3003	Non-metallic materials - Structural adhesives - Determination of mass per unit area of film adhesives - Test method
ISO 7253	Paints and varnishes - Determination of resistance to neutral salt spray

4 Technical requirements

4.1 General

Specification values are minimum except where stated.

4.2 Specific

4.2.1 General requiremnets

Unless otherwise stated all test values specified are minium average.

4.2.2 Material description

See Table 1.

Table 1: Material Description

Characteristic number	Characteristic	Requirements		
1	Material description	Structural adhesive film		
2	Formulation	Adhesive: epoxy based Carrier: organic polymer		
3	Technical specification	AIMS10-01-000 Part 1 AIMS10-01-000 Part 3		
4	Storage conditions	< -18°C		
5	Storage	12 months		
6	Grade	A	B	C
7	Mass per unit area (g/m ²)	150	250	300
8	Carrier type	Non woven scrim or knitted		

4.2.3 Worklife and storage conditions

See Table 2.

Table 2: Requirements for storage and work life of the adhesive film

Storage life at -18°C [days]	Work life at 28 ± 2 °C 70 ± 5% r.h. [days]	Test method	Test temp. [°C]	Symbol	Unit	Requirements
< 3	0 15 ²⁾ 30	Surface tack	-	-	-	Reference ¹⁾
		HPLC (AITM 3.0001)	-	P _X /P _R P _Y /P _R	-	Same peak area ratios as reference ¹⁾
		DSC (AITM3-0002)	-	T _g	°C	Change ≤ 5°C compared with reference ¹⁾
		Single lap shear (metal/metal) (EN 2243-1 Type 1)	23 ± 2 120 ± 2	Ô	MPa	Same strength as reference ¹⁾
		Flow (EN 2243-7)	-	FI	%	³⁾
360	0 15 ²⁾ 30	Surface tack	-	-	-	reference ¹⁾
		HPLC (AITM 3.0001)	-	P _X /P _R P _Y /P _R	-	Same peak area ratios as reference ¹⁾
		DSC (AITM 3.0002)	-	T _g	°C	Change ≤ 5°C compared with reference ¹⁾
		Single lap shear (metal/metal) (EN 2243-1 Type 1)	23 ± 2 120 ± 2	Ô	MPa	Reference ¹⁾
		Flow (EN 2243-7)	-	FI	%	³⁾
¹⁾ Reference: value obtained when testing the adhesive film with less than three days of storage time and 0 days of work time. ²⁾ Unless an alternative period is agreed between the manufacturer and Airbus MAAP at the start of qualification. ³⁾ To be determined during qualification testing and recorded in the IPS						

4.2.4 Processing conditions

See Table 3.

Table 3: Processing conditions

No.	Characteristic	Unit	Requirements	
			Parts manufacturing	Qualification
1	Cure equipment	-	Autoclave	Autoclave
2	Heating rate	°C/min	0,2 to 5	2 ± 1
3	Cure temperature	°C	170 to 180	175 ± 5
4	Cure time	Min	60 to 135	90 ± 15
5	Bonding pressure	kPa	100 to 1000	320 ± 25
6	Cooling rate	°C/min	< 3	2 ± 0,5
7	Vacuum (pressure) ¹⁾	kPa	5 to 100	30 ± 5
¹⁾ Vent to atmosphere when autoclave pressure reaches 140 kPa				

4.2.5 Requirements for uncured adhesive

See Table 4.

Table 4: Requirements for uncured adhesive film

Test No.	Property	Test Method	Symbol	Unit	Requirements		
1	Mass per unit area	EN3003	Am	g/m ²	Grade A	Grade B	Grade C
					150 ± 25	250 ± 25	300 ± 25
2	Flow	EN2243-7	Fl	%	≥ 150	≥ 200	≥ 200
3	Volatile content	EN 2558	Vc	%	≤ 1		
4	DSC (curing chracteristics)	AITM 3.0002	T _{os}	°C	- ¹⁾		
			T _p	°C			
			ΔH	J/g			
			ΔH ₁₀₀	J/g			
			Tg	°C			
5	HPLC (resin composition)	AITM 3.0001	P _x /P _r P _y /P _r (peak area ratios)	-	- ^{1) 2)}		
6	IR (analysis of organic compounds)	AITM 3.0003 (semi quantitative analysis)	i _E , i _X , i _Y	-	- ¹⁾		

¹⁾ To be determined during qualification testing and recorded in the IPS

²⁾ Details of the carrier shall be recorded in the IPS

4.2.6 Requirements for cured adhesive

Dry: See Table 5.

Conditioned(Wet): See Table 6.

Resistance to fluids: See Table 7

Co-cured and co-bonded: see Table 8

Table 5: Requirements for the adhesive film in cured condition (dry specimens)

Test No.	Property	Test Method	Test Temp. [°C]	Symbols	Units	Requirements	
						Grade A	Grade B and C
1a	Single lap shear strength (composite/composite)	AITM1-0019 Specimen S7	-55	σ̂	MPa	16,5	
			23			17,5	
			80			17,5	
			120			8,0	
1b	Double lap shear strength (composite/composite)	AITM1-0019 Specimen D13	-55	σ̂	MPa	24,0	
			23			24,0	
			80			24,0	
			120			10,0	
2	Single lap shear strength (metal/metal)	EN2243-1 (Type 1)	23	σ̂	MPa	24,0	
			80			19,5	
3	Shear stress and shear strain behaviour (metal/metal)	EN2243-6	23	σ̂ _B α̃ _B G _e	MPa % MPa	1)	
			80				
4	Peel load (metal/metal)	EN2243-2	23	FI	N	65	
5	Peel load (metal/honeycomb core)	EN2243-3	23	FI	N	NA	110
6	Flatwise tensile strength (metal/honeycomb core)	EN2243-4	23	R	MPa	NA	4,5
			80				4,5
			120				3,5
7	Glass transition temperature	AITM1-0003	-	Tg _{peak}	°C	1)	
				Tg _{onset}		135	
1) To be determined during qualification testing and recorded in the IPS							

Table 6: Requirements for the adhesive film in cured and condition (wet specimens)

Test No.	Property	Test Method	Conditionings	Test Temp [°C]	Symbol	Units	Requirements
1a	Single lap shear strength (composite/composite)	AITM1-0019 (S7)	70±2°C at 85% RH for 2000 h	23	Ô	MPa	15,5
				80			13,5
1b	Double lap shear strength (composite/composite)	AITM1-0019 (D13)		23	Ô	MPa	16,5
				80			13,5
3a	Glass transition temperature	AITM1-0003		-	T _{g_{peak}}	°C	- ⁽¹⁾
					T _{g_{onset}}		95
2a	Single lap shear strength (composite/composite)	AITM1-0019 (S7)	70±2°C in water for 2000 h	23	Ô	MPa	16,5
				80			13,5
2b	Double lap shear strength (composite/composite)	AITM1-0019 (D13)		23	Ô	MPa	16,5
				80			13,5
3b	Glass transition temperature	AITM1-0003		-	T _{g_{peak}}	°C	- ⁽¹⁾
					T _{g_{onset}}	°C	90
1) To be determined and reported in the IPS							

Table 7: Requirements for adhesive film in cured conditions (resistance to fluids)

Test No.	Property	Test Method	Conditioning	Test T° [°C]	Symbol	Units	Requirements
1	Single lap shear strength (metal/metal)	EN2243-1 Type 1	70±2°C in hydraulic fluid for 2000 h	23	Ô	MPa	20,0
				80			15,0
2	Single lap shear strength (metal/metal)	EN2243-1 Type 1	50±2°C in hydraulic fluid/water mixture for 2000 h	23	Ô	MPa	20,0
				80			15,0
3	Single lap shear strength (metal/metal)	EN2243-1 Type 1	23±2°C in fuel for 2000 h	23	Ô	MPa	20,0
				80			15,0
4	Single lap shear strength (metal/metal)	EN2243-1 Type 1	23±2°C in solvent for 24 h	23	Ô	MPa	22,0
				80			19,0
5	Peel load (metal/metal)	EN2243-2	Exposure at 23°C to phenolic paint stripper for 24 h	23	F	N	65
6	Peel load (metal/metal)	EN2243-2	Exposure at 23°C to non-phenolic paint stripper for 24 h	23	F	N	65

Table 8: Requirements for adhesive film in cured conditions (co-cured and co-bonded specimens)

Test No.	Property	Test Method	Conditionings	Test T° [°C]	Symbol	Units	Requirements
1	Compatibility-Glass transition temperatures	AITM1-0003	Dry	-55 to > T _g	-	-	No change on the thermal transitions of adhesive and composite shall be observed due to the co-curing process
2			70±2°C at 85% RH for 2000 h				
3			70±2°C in water for 2000 h				
4	Flatwise tensile strength (CF skin/glass honeycomb)	AITM1-0025 and section 4.3.5 of part 1	Dry	23	Ó _B	MPa	5,8
				80			4,5
5			70±2°C at 85% RH for 2000 h	23	Ó _B	MPa	5,3
				80			3,0
6			70±2°C in water for 2000 h	23	Ó _B	MPa	5,3
				80			3,0

4.2.7 Requirements for the cure cycle

See Table 9.

Table 9: Requirements of the cure cycle for adhesive films

Curing parameters	Test conditions	Test Method	Test temp. (°C)	Symbol	Unit	Requirements
Heat up rate	0,2°C/min	Single lap shear (comp./comp.) (AITM1-0019 S7)	23	Ô	MPa	Reference
		Peel (metal/metal) (EN2243-1 Type 1)	23	F	N	Reference
Time/temperature	50 min/160°C (2°C/min)	DSC (AITM3-0008)	23	ÃH ₁₀₀	(J/g)	Reference
Bonding pressure	100 kPa 1000 kPa	Single lap shear (metal/metal) (EN2243-1 Type 1)	23	Ô	MPa	Reference
	100 kPa 1000 kPa	Single lap shear (comp./comp.) (AITM1-0019 S7)	23	Ô	MPa	Reference
Multiple curing	5 cycles at 180°C/120 min	Single lap shear (metal/metal) (EN2243-1 Type 1)	23	Ô	MPa	Reference
		Peel (metal/metal) (EN2243-2)	23	F	N	Reference

5 Designation

See relevant ABS.

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS10-01-000.

7 Incoming Inspection

Material qualified against this AIMS shall undergo “Technical Incoming Inspection” as per AIMS10-01-000 Part 100 prior to use.

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1		New standard
2 10/99		
3 04/15	- - - Chapter 2 Chapter 7	Updated to latest template Revision sheet added Latest standardization rules applied AIMS10-01-000 part 100 added All referenced documents added Chapter 7 defined in line with Incoming Inspection (Part 100) implementation